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Leveraging Data Acquisition and Analysis to Optimize Hydraulic Fracturing and Well Testing in a Tight Gas-Condensate Greenfield in North Central Oman

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Abstract

This paper presents a comprehensive, data-driven approach to optimizing hydraulic fracturing and well testing in a tight gas-condensate greenfield in North Central Oman. By integrating multi-disciplinary datasets, including well logs, production data, fracture diagnostics, and analytical techniques such as Pressure Transient Analysis (PTA) and Diagnostic Fracture Injection Tests (DFITs), the study enabled iterative design improvements across three well campaigns. The approach emphasized real-time feedback loops and continuous optimization, resulting in a 49% improvement in gas rate per proppant tonnage, a 64% reduction in flowback duration, and a 50% reduction in PLT survey time.

Key insights included the shift from volume-driven to efficiency-driven stimulation strategies, the successful reduction of fracture stages without compromising reservoir contact, and the selective application of DFITs based on diagnostic evidence of stage communication. These optimizations led to enhanced fracture efficiency, reduced material usage, and improved operational performance.

The findings demonstrate the value of integrating diagnostics and analytics early in field development to reduce uncertainty, improve productivity, and lower costs. This methodology offers a scalable framework for sustainable development in similarly complex tight gas-condensate reservoirs.

Introduction

Shell Development Oman (SDO) currently operates deep tight gas-condensate reservoirs in Blocks 10 & 11 within North-Central Oman (see [Fig. 1](#) below) targeting the Barik, Al-Bashair and Miqrat clastic reservoirs which constitute the lower part of the Haima Supergroup ([Al Amri et al., 2024](#)). The Northern Oman Salt Basins contain some of the largest gas deposits within Oman ([Al-Jahdhami et al., 2021](#); Droste, 1997). The Haima Supergroup of Oman consists of predominantly siliciclastic sediments of Cambrian to Early Silurian age (Droste, 1997). As highlighted in [Fig. 2](#), the reservoirs discussed in this paper are characterized by multi-stacked sand body intervals interbedded with shales with porosities of 4-8% and a permeability range of 0.1-5 md.

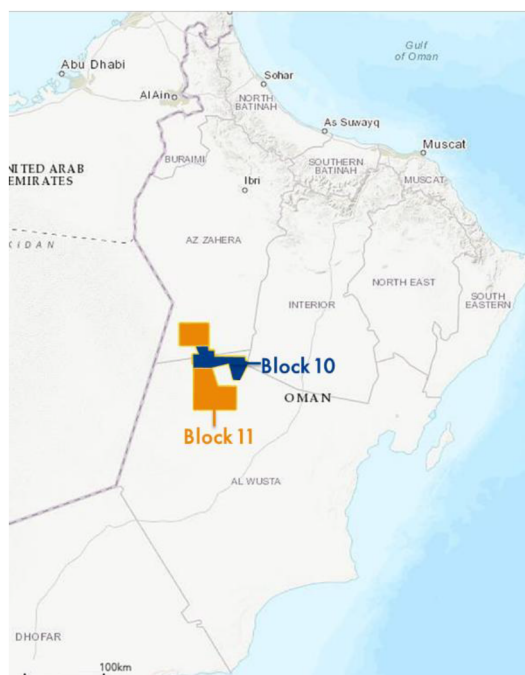


Figure 1—Map of Shell's operated concession areas (Blocks 10 and 11) within North-Central Oman

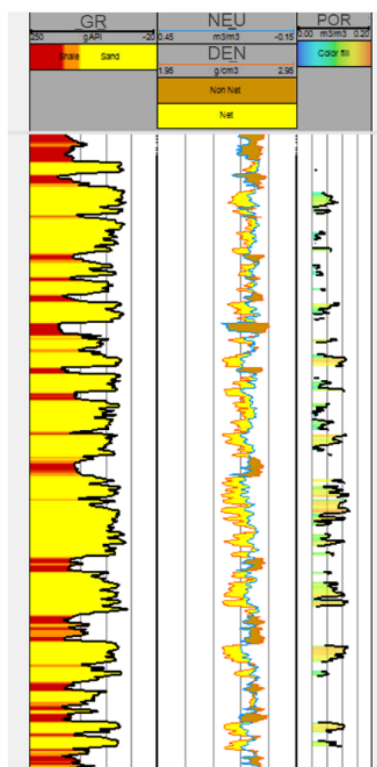


Figure 2—Example of a well log showing the sand-shale sequences within the Lower Haima reservoirs

Development of deep tight gas-condensate reservoirs presents significant challenges within the industry due to their low permeability, complex fluid dynamics, and high uncertainty in reservoir performance. To maximize hydrocarbon recovery and commercially produce such reservoirs, wells are hydraulically fractured. Hydraulic fracturing is a key technology for stimulating low-permeability reservoirs, but its success relies heavily on understanding key subsurface parameters, and optimizing fracture design to

improve fracture effectiveness, enhance well deliverability, and reduce operational costs (Al Shibani et al. 2025).

In North Central Oman, tight gas-condensate reservoirs posed unique challenges, necessitating an integrated strategy to improve fracture effectiveness, enhance well deliverability, and reduce operational costs. Traditional approaches to fracturing often rely on simplistic models and pre-determined designs, which may not account for dynamic changes in reservoir behavior and well performance. This study aims to bridge this gap by leveraging comprehensive data acquisition, analytical techniques, and real-time performance monitoring to drive continuous optimization.

The methodology presented in this paper integrates multiple datasets, including well logs, early production data, pressure transient analysis (PTA), and Diagnostic Fracture Injection Test (DFIT) results, to refine hydraulic fracturing and well testing workflows. By continuously analyzing pre-drill well campaign data (wells drilled and tested before production facility start-up) through to First Gas and integrating it with post-fracture production responses, a dynamic workflow was developed that allowed for iterative design improvements. This approach not only enhanced fracture conductivity and production rates but also led to significant reductions in operational inefficiencies and costs.

This paper focuses on Block 10 which is currently in the development phase and provides a detailed case study of how a data-driven approach was successfully applied in a tight gas-condensate greenfield in Oman. It demonstrates how integrating well and reservoir data with real-time fracturing performance metrics enabled more precise decision-making, leading to a reduction in operational costs while enhancing well productivity. The findings from this study offer valuable insights for engineers and field operators working in similar challenging reservoir environments.

Methodology and Data Integration Approach

This study adopted a comprehensive, multi-disciplinary methodology to optimize hydraulic fracturing and well performance in tight gas-condensate reservoirs. The approach was built on the integration of diverse data sources and advanced analytical techniques, enabling a dynamic, iterative optimization process that spanned from pre-drill well campaigns through to First Gas.

Data Sources and Acquisition Strategy

At the core of this methodology was a robust data acquisition framework designed to capture both static and dynamic subsurface information throughout the well lifecycle. Key data sources included:

- Well and reservoir data – such as geological and petrophysical logs, core analysis, and pressure measurements.
- Production test data – including flowback responses from post-frac well tests, pressure build-ups, and early well-life flowrates.
- Fracture modeling results – derived from both empirical and numerical models calibrated with field-specific rock and fluid properties.

A strong emphasis on data acquisition was placed from the outset of the development program. The process began during the pre-drill phase, ensuring that critical subsurface insights were available to inform initial fracture design decisions. This effort continued seamlessly through the drilling, completion, and testing stages, ahead of First Gas. By embedding data collection throughout the well lifecycle, the team established a continuous feedback loop – enabling real-time design adjustments based on evolving reservoir understanding and well performance. This iterative, end-to-end acquisition strategy played a key role in refining stimulation designs and maximizing the value of each subsequent well.

Analytical Techniques and Integration Workflow

To transform raw data into actionable insights, the study employed a suite of fundamental analytical techniques. The team conducted in-field benchmarking to assess well performance across campaigns. This involved continuously updating cross-plots deriving correlations between petrophysical properties, well performance, and frac job parameters as shown in Fig. 3 below. By linking rock quality indicators, such as Equivalent Gas Column (EGC or $S_g \cdot \phi \cdot H$), with production outcomes and treatment parameters, the team developed a more predictive understanding of how to tailor fracture designs to specific reservoir conditions. This enabled identifying variances from data trends and refining the definition of what "good" looks like in terms of well productivity and stimulation efficiency. By analyzing deviations from expected performance, the team was able to identify underperforming wells and investigate root causes.

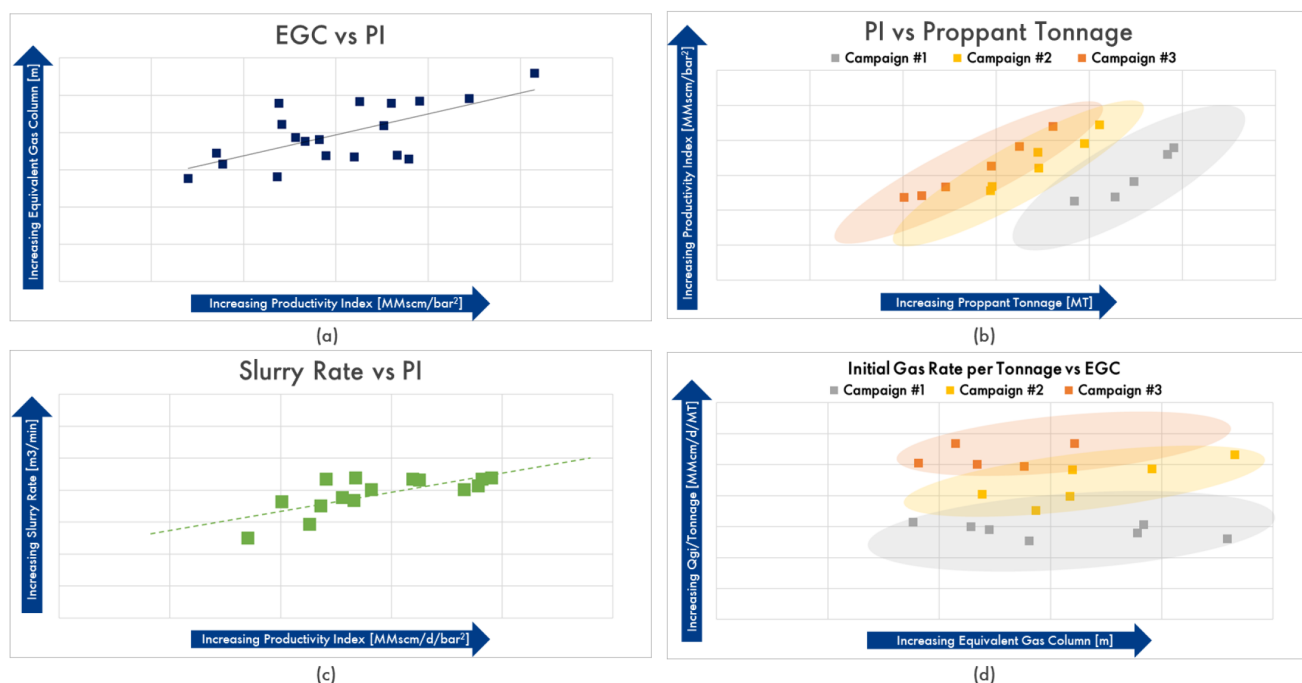


Figure 3—Examples of cross-plots used to correlate between well petrophysical properties & performance data, and frac job parameters; a) Well equivalent gas column vs estimated average well Productivity Index (PI) used to identify dependency between productivity and rock quality; b) Well Productivity Index plotted against pumped Proppant Tonnage for each well campaign to highlight improvement in productivity with optimization efforts; c) Frac job average slurry rate pumped per well vs Productivity Index to identify dependency between productivity and key frac parameters; d) Initial gas rate per pumped tonnage per well plotted against equivalent gas column for each well campaign to highlight improvement in fracture efficiency with optimization efforts

In addition to these core techniques, Pressure Transient Analysis (PTA) was used to evaluate reservoir properties, fracture effectiveness, and well deliverability (discussed more in follow-up sections). It provided key indicators such as flow regimes, permeability estimates, and effective fracture half-lengths. In parallel, Diagnostic Fracture Injection Tests (DFITs) were conducted prior to fracturing stages to estimate in-situ stress, fracture propagation pressure (Process zone stress), closure pressure, and leak-off behavior—critical inputs for fracture model calibration (Mullen et al, 2010). These analyses were integrated into a centralized workflow that combined static and dynamic data to continuously update the understanding of reservoir behavior and stimulation effectiveness. Fracture modeling outputs were iteratively refined using field data, while production test results were benchmarked against modeled expectations to validate design assumptions.

The integration of these data streams enabled a closed-loop optimization process, where each well informed the next. This iterative approach allowed for continuous refinement of fracture design, stage count,

and treatment parameters, ensuring that each subsequent well benefited from the learnings of the previous one.

Fracture Modeling and Calibration

Fracture modeling played a pivotal role in this workflow by integrating diagnostic, petrophysical, and geomechanics datasets to better characterize the reservoir and predict fracture behavior. The process began with the interpretation of DFIT results, which provided foundational inputs such as fracture propagation pressure, fracture closure pressure, and pore pressure. These parameters were used to calibrate the geomechanics and main fracture model.

A key step in the modeling process was pressure matching the DFIT results to ensure that the simulated fracture behavior accurately reflected the reservoir's response to fluid injection as illustrated in Fig. 4 (Mullen et al, 2010). This calibration increased confidence in the model's predictive capability and allowed for scenario testing prior to executing the main frac job. Petrophysical logs along with core data – including Gamma ray, Resistivity, Neutron-density, and Sonic logs – were used to derive rock properties such as porosity, permeability, Young's modulus, and Poisson's ratio. These properties were spatially distributed across the reservoir to construct a detailed geomechanics model and as an input to fracture design. The model enabled the identification of stress contrasts, mechanical stratigraphy, and potential fracture barriers, which are essential for predicting fracture height growth and containment.

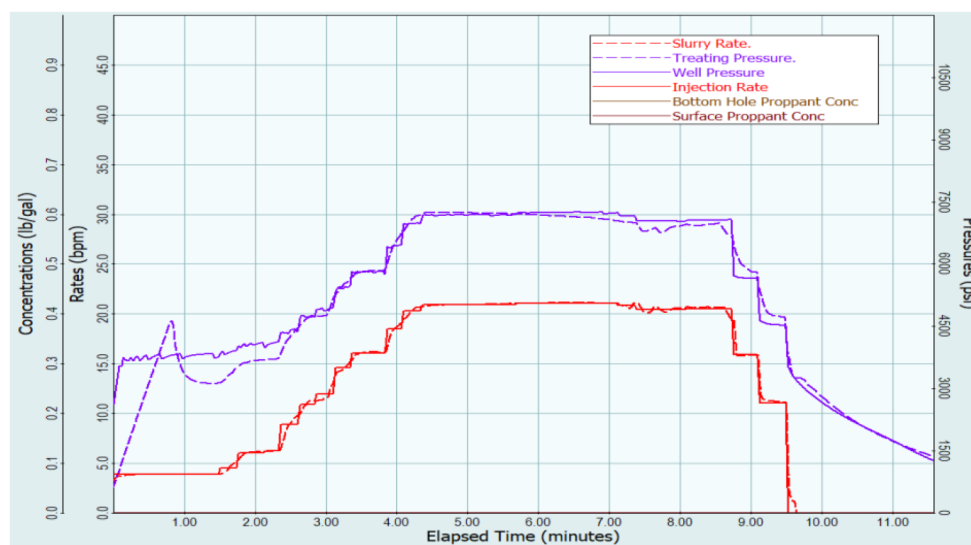


Figure 4—Example of Pressure Matching of actual DFIT treatment schedule with a fracture model

The modeling also allowed for testing fracture height growth under different scenarios, particularly in tight gas reservoirs where the presence of baffles or stress barriers can significantly influence vertical and lateral fracture propagation. This was critical in avoiding fracture growth into water-bearing zones and optimizing lateral fracture extension within the target pay zone. By simulating these scenarios, the team was able to refine stage spacing, fluid volumes, and proppant concentration to maximize reservoir contact while minimizing operational risks. To validate the modeled fracture geometry, a radioactive tracer was pumped with each frac stage followed by a gamma ray logging run post-fracturing. The tracer data provided direct measurements of proppant placement and vertical fracture extent. As illustrated in Fig. 5, the tracer log confirmed that fracture growth was successfully contained within the main target interval, aligning closely with the modeled predictions. This validation was crucial in confirming that the fracture design effectively sweet-spotted the gas column, avoided water zones, and maximized hydrocarbon recovery.

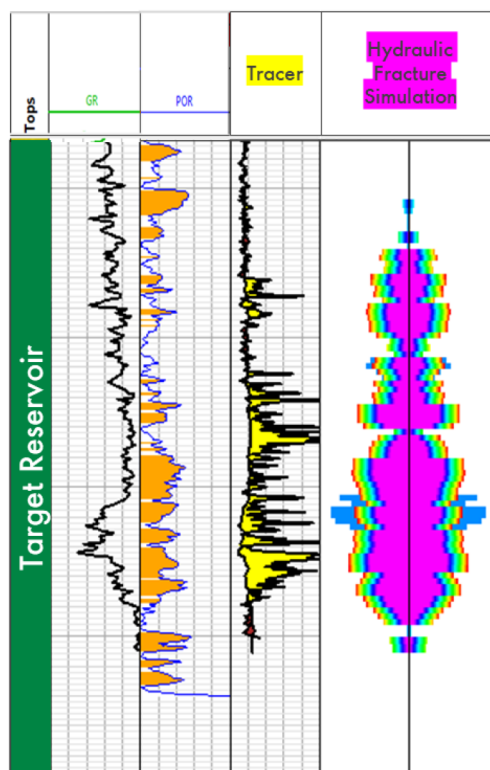


Figure 5—Well Log showing a radioactive tracer extending across the target reservoir matching the predicted hydraulic fracture growth simulated from the fracture model

Hydraulic Fracturing Optimization Strategy

Frac intensity optimization

The hydraulic fracturing optimization campaign was kicked-off with a clear objective: to enhance reservoir stimulation efficiency while optimizing material usage and operational costs. Over the course of 3 well campaigns, the team implemented a structured optimization strategy that relied on continuous data acquisition, diagnostic interpretation, and iterative design refinement.

The evaluation was focused on the total treatment per well across multiple stimulation campaigns. This evolution is clearly illustrated in Fig. 6, which shows the change in performance and design across the well campaigns in terms of normalized initial rate per proppant tonnage (green bars), average proppant intensity (orange line) and average treatment slurry rate (yellow line). As the campaign progressed, proppant intensity and tonnage were optimized, yet initial gas rates remained stable or improved across similar reservoir quality, indicating performance gains were largely due to improved stimulation. This reflects a successful transition from a volume-driven to a value-driven stimulation strategy, where optimized design and real-time feedback enabled both technical and commercial gains.

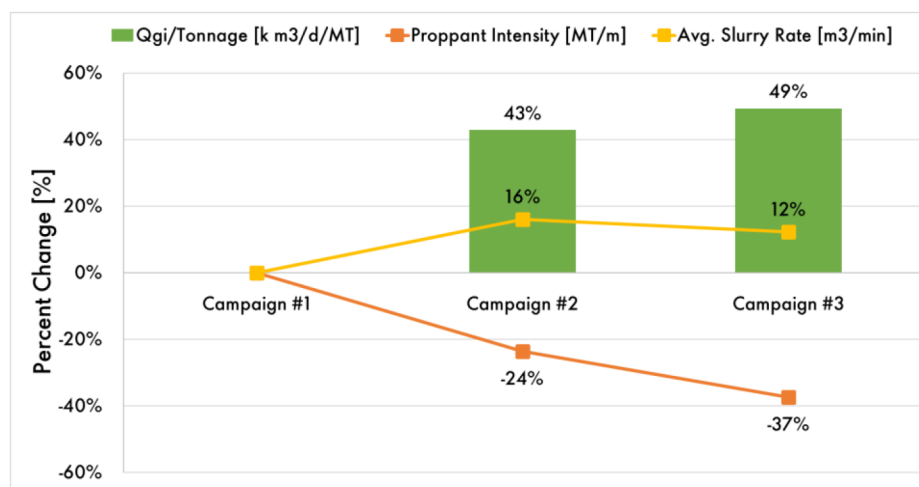


Figure 6—Evolution of hydraulic fracturing efficiency across the three well campaigns. The green bars represent normalized initial gas rate per ton of proppant, highlighting a steady improvement in stimulation effectiveness. The Proppant Intensity (orange line) indicates a significant reduction in proppant usage per net pay, while the Average Slurry Rate (yellow line) reflects adjustments in fluid pumping strategy. The combined trends highlight the impact of iterative design optimizations in enhancing fracture efficiency and reducing material input.

Campaign #1 marked the initial phase of the hydraulic fracturing optimization effort. Wells were completed using designs characterized by high proppant tonnage and moderate treatment intensity. These early completions served as a baseline, offering critical insights into fracture performance, and reservoir response. While initial gas rates were within acceptable limits, the relatively high material consumption and variability in stimulation effectiveness revealed inefficiencies in the design. These observations underscored the need for a more targeted approach to improve both technical performance and cost-effectiveness.

Informed by learning and data acquired – particularly from Diagnostic Fracture Injection Tests (DFITs), Pressure Transient Analysis (PTA), and production testing - Campaign #2 introduced a more refined design strategy. The team implemented progressive changes to perforation & stage spacing, proppant schedules, and fluid systems, aiming to enhance fracture coverage and conductivity while continuously monitoring their impact. During this phase, proppant intensity was reduced by 24%, while the average slurry rate increased by 16%, reflecting a shift toward more efficient fluid placement. Notably, the normalized initial gas rate per proppant tonnage improved by 43%, indicating a significant gain in stimulation efficiency.

A key insight emerged during this phase: proppant intensity could be reduced without compromising initial gas rates. In fact, several wells demonstrated improved performance despite using less proppant per well net pay. This indicated that fracture efficiency, rather than total volume, was the primary driver of well productivity. The EGC remained relatively stable across wells, reinforcing that the observed improvements were largely attributable to design optimization rather than geological variability.

Further supporting this observation, PTA tests conducted in several wells showed no significant improvement in dimensionless fracture conductivity in wells where high proppant tonnage was pumped. The example shown in Fig. 7 below compares two wells with similar reservoir properties and completion designs where Well-B (blue) was treated with ~40% less proppant tonnage compared to Well-A (Red). This finding suggested that simply increasing proppant volume did not necessarily translate to greater fracture extension or improved reservoir contact. Instead, it highlighted the opportunity to optimize proppant usage without compromising stimulation effectiveness.

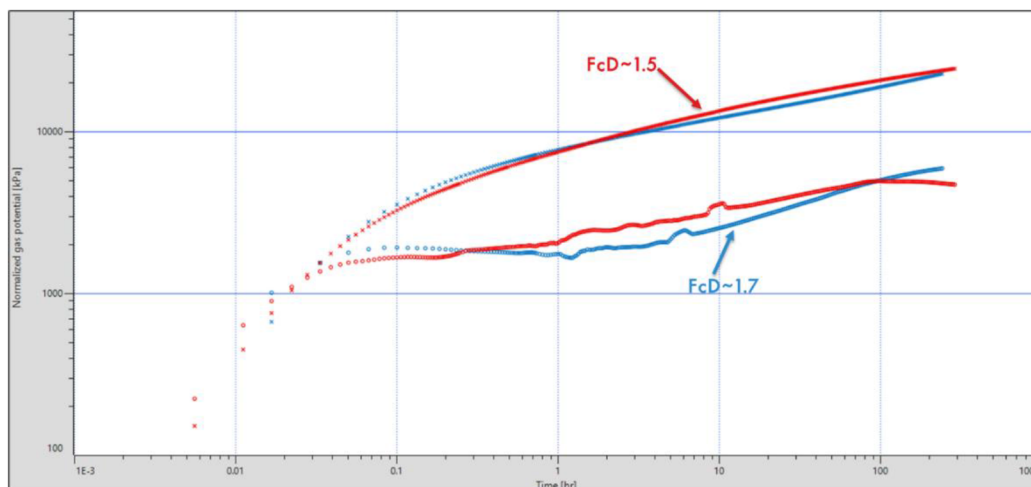


Figure 7—Log-log plot of pressure build-up and derivative curves showing Well-B (Blue) treated with a proppant tonnage ~40% less than in Well-A (Red)

By Campaign #3, the fracture designs had reached a mature and optimized state. The final wells delivered gas rates that were comparable to, or in some cases exceeded, those of earlier wells, despite utilizing significantly lower proppant volumes. Specifically, proppant intensity was reduced by 37%, average slurry rate increased by 12%, and gas rate per proppant tonnage improved by 49% compared to Campaign 1. While the most substantial gains were realized between Campaigns 1 and 2, Campaign 3 achieved a marginal yet meaningful improvement over Campaign 2, further enhancing overall efficiency. A critical aspect of this optimization phase was the deliberate effort to avoid fracture growth into water-bearing zones while maximizing lateral fracture extension within the target pay zone. This contributed to sustaining production performance and minimizing water production risks.

Further evidence of this optimization is presented in Fig. 8, which tracks the change in Frac fluid & materials cost and quantity per stage over the 3 campaigns. The chart shows a consistent reduction trend in fluid and material cost per stage, decreasing by 17% by Campaign #3. This reduction aligns with the implementation of optimized designs and operational efficiencies, confirming that the campaign not only improved technical outcomes but also delivered measurable commercial benefits. Similarly, the trend in fluid volume and proppant tonnage per frac stage over the campaigns show a clear declining trend, with fluid volume per stage decreasing by 28%, and proppant tonnage per stage by 24%. These reductions reflect the campaign's success in refining treatment designs to be more efficient, targeted, and cost-effective, without sacrificing well performance.

Collectively, these advancements translated into notable cost reductions and provided strong validation for the data-driven, adaptive design approach applied throughout the campaign. Together, the results demonstrate that a diagnostic-informed, efficiency-focused approach to hydraulic fracturing can unlock substantial value in tight gas-condensate reservoirs. By focusing on fracture efficiency rather than total volume, the team consistently delivered strong production performance while reducing material usage and operational costs.

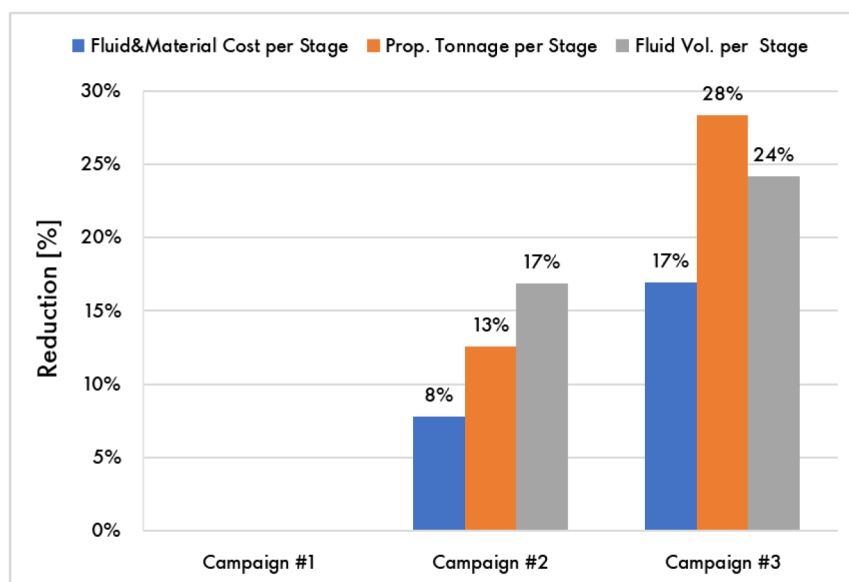


Figure 8—Percent reduction across hydraulic fracturing campaigns in Cost per Stage (blue), Proppant Tonnage per Stage (orange), and Fluid Volume per Stage (grey)

Frac Stage Optimization

As part of the broader hydraulic fracturing optimization strategy, a focused effort was made to evaluate and refine the number of fracture stages in the Miqrar reservoir. This initiative aimed to balance stimulation effectiveness with operational efficiency and cost control, particularly considering evolving insights from field diagnostics and production performance.

As shown in Fig. 9, the Miqrar reservoir development has undergone a clear transformation in completion design across three campaigns. The chart illustrates a steady reduction in the average number of fracture stages per well accompanied by significant increases in both proppant tonnage and slurry rate per stage.

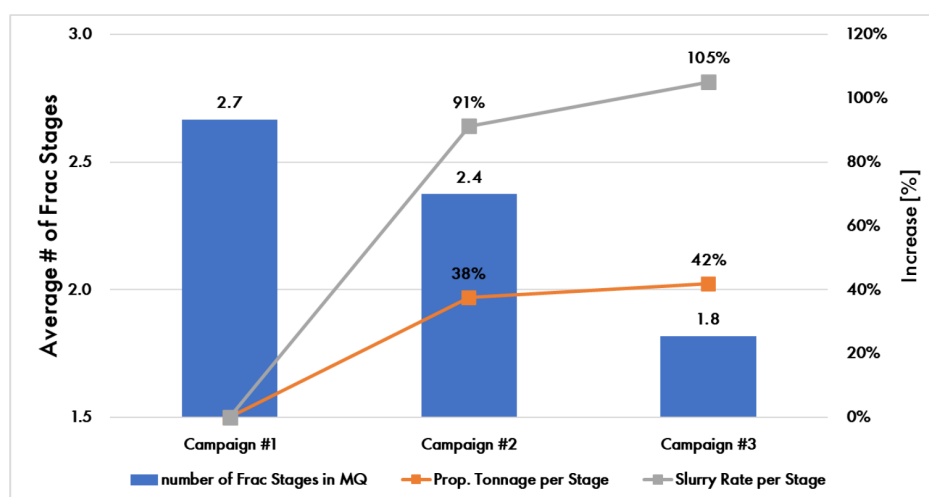


Figure 9—Evolution of average fracture stages, proppant tonnage, and slurry rate per stage across development campaigns

This evolution reflects a strategic shift from a conservative, high-stage-count design to a more efficient, high-intensity stimulation strategy:

- Campaign #1: Initiated with an average of 2.7 stages per well, based on early geological interpretations and cautious assumptions about reservoir heterogeneity.

- Campaign #2: Reduced to an average of 2.4 stages per well, with a ~38% increase in proppant and a 91% increase in slurry rate per stage. This marked a pivot toward maximizing individual stage effectiveness while maintaining production performance and reducing complexity.
- Campaign #3: Further reduced to an average of 1.8 stages per well, with proppant per stage up by ~40% and slurry rate more than doubling compared to the initial campaign. This approach emphasized fewer, more effectively stimulated stages, leveraging improved understanding of fracture propagation and reservoir connectivity

Together, these changes demonstrate a data-driven, technically grounded approach to completion optimization. By simplifying stage count and intensifying treatment per stage, the team maintained or improved well performance, streamlined operations, and significantly reduced completion costs.

To support this optimization, a suite of diagnostic tools was deployed, including radioactive tracers, fracture modeling, and production logging. These techniques provided critical insights into fracture height propagation, stage coverage, and reservoir contribution. The data revealed that in many cases, fractures were extending vertically beyond the intended zone, suggesting that fewer, more effectively placed stages could achieve the same or better reservoir contact. Specifically, if fracturing size is too large, the created fracture may grow into the water zone, which would result in compromising gas production. By reducing water production in wet or transition zones, better well productivity can be realized.

Fig. 10 shows an example of two wells targeting the same reservoir units R1 & R2. Tracks from left to right are for GR, Neutron & Density, Porosity, Perforated zones with Stage number and last, the radioactive tracer logs. Well-A on the left was perforated and stimulated in two stages targeting both reservoir units R1 and R2 with distinct radioactive tracers pumped in each stage. Well-B on the right was stimulated using a single stage covering both reservoir units. After running the tracer logs, both wells showed indications of extensive fracture growth covering the target reservoir units. Well-B even showed a single frac stage was sufficient to stimulate both reservoir units.

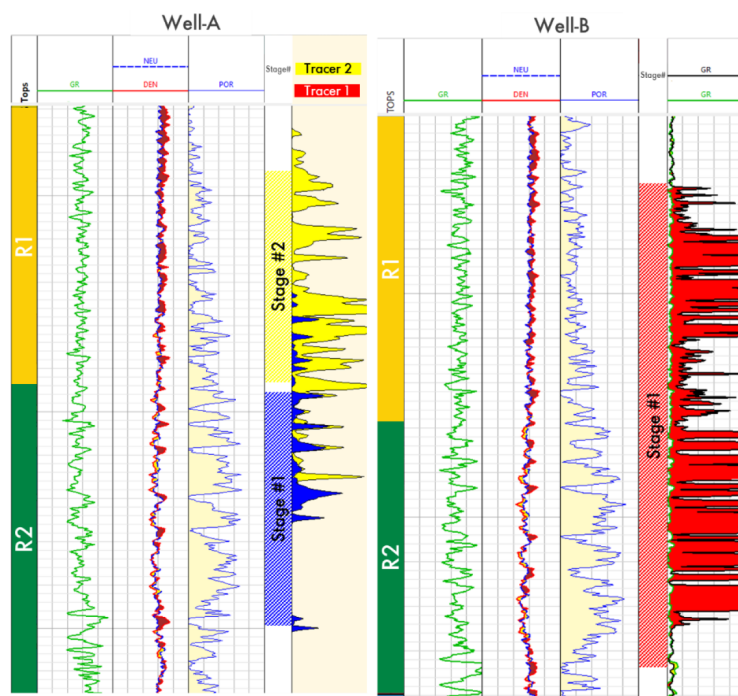


Figure 10—Petrophysical logs of two wells (A and B) showing distinct radioactive tracers pumped per stage. The radioactive tracer propagation measured in both wells indicates sufficient fracture coverage across the target reservoirs was achieved from 1-2 frac stages

DFIT optimization

As part of the continuous effort to streamline hydraulic fracturing operations and reduce non-productive time, the team undertook a focused review of Diagnostic Fracture Injection Tests (DFITs) across the campaign. DFITs were routinely conducted before each fracture stage to estimate key subsurface parameters such as pore pressure, formation breakdown pressure, fracture closure pressure, fracture propagation pressure, and leak-off behavior via fluid efficiency (Mullen et al, 2010; McClure et al, 2019). These tests are essential for understanding the mechanical response of the reservoir and for calibrating fracture models, which in turn support the design optimization to deliver effective and efficient hydraulic fracturing treatments.

In a hydraulic fracturing operation involving two stages, each stage is preceded by a DFIT. After completing Stage 1, a bridge plug was set to isolate the zone before perforating and treating Stage 2. The DFIT involves injecting a small volume of fluid into the reservoir at a controlled rate and then shutting in the well to monitor the pressure decline (fall-off) over time. The pressure response provides insights into fracture geometry, reservoir permeability, and fluid leak-off characteristics (Mullen et al, 2010; McClure et al, 2019). These data are critical for refining stimulation designs, particularly in tight and heterogeneous formations.

However, analysis of DFIT pumping pressure profiles during the campaign revealed a recurring anomaly: No clear indication of formation breakdown pressure in the second frac stage Fig. 11(a). In a typical tight rock DFIT, pressure is expected to build gradually until a clear inflection point indicating formation breakdown and fracture initiation as shown in Fig. 11(b). The absence of a clear breakdown in Fig. 11(a) suggested that the test interacted with a pre-existing fracture network.

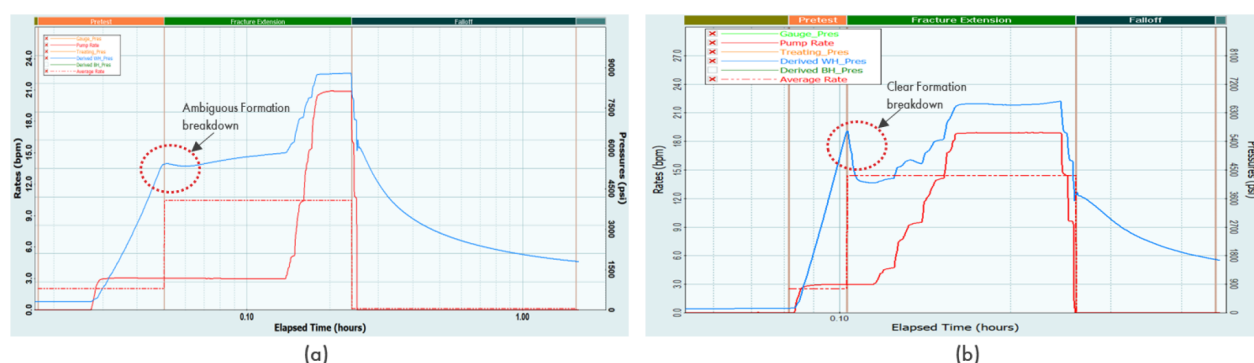


Figure 11—DFIT analysis plots of rate and pressure vs elapsed time showing distinct pressure behaviors; a) an example where the pressure build-up does not show a clear indication of formation breakdown indicating likelihood of connecting to the fracture network from the previous frac stage; b) an example where the pressure build-up shows a clear indication of formation breakdown

Further investigation confirmed that this pressure drop was likely due to interaction with a previously pumped fracture stage, even though the wellbore had been positively isolated with a bridge plug. In these cases, the DFIT interval was apparently influenced by the earlier stimulation, via fracture height growth in the reservoir. This was strongly supported by radioactive tracer results from a sample of wells across the field as shown in Fig. 12 which indicate fracture height extending from the bottom stage into the upper stage where the subsequent DFIT was targeted. The bridge plug had been found intact when milling, and the tracer data validated that the fracture network was not contained within the target zone, confirming communication between stages.

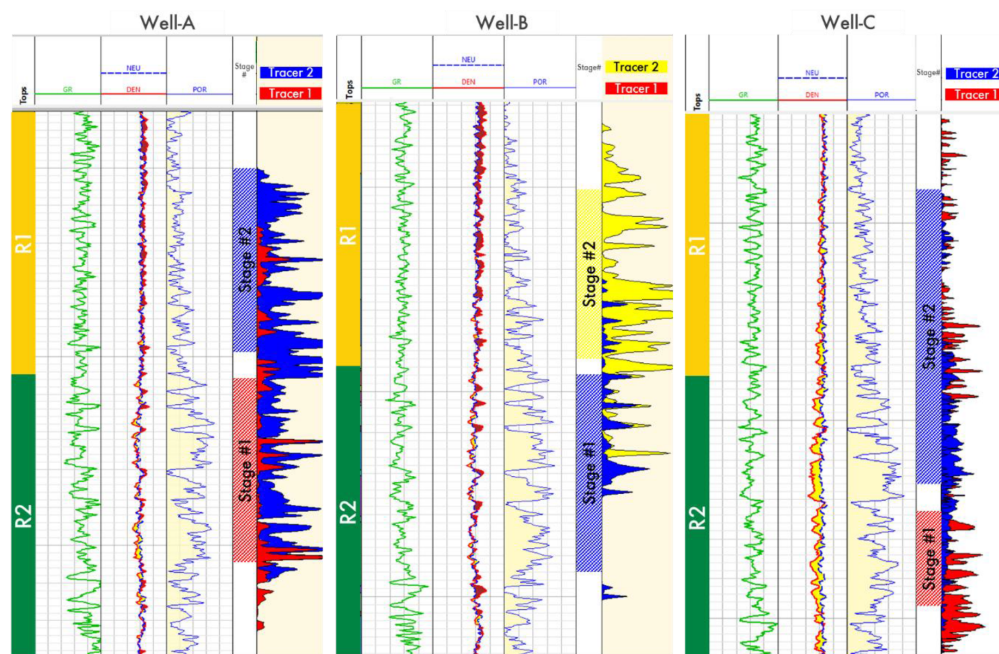


Figure 12—Petrophysical logs of 3 wells (A, B and C) showing distinct radioactive tracers pumped per stage. The radioactive tracer propagation measured in all wells indicates fracture growth from Stage#1 in R2 to R1 resulting in fractures initiated from the DFIT in Stage#2 to connect to the fracture network extending from Stage#1 leading to non-representative DFIT outcomes.

These findings had two major implications. First, they highlighted that DFIT results in such scenarios were not representative of the true reservoir conditions, and therefore could not be reliably used to calibrate fracture models. Second, and more importantly, they enabled the team to proactively identify and eliminate DFITs in stages where similar conditions were expected. This decision, informed by fracture modeling, tracer diagnostics, and operational experience, led to significant savings in operational time and cost.

Despite the reduction in DFIT frequency, the tests that were conducted in isolated and representative intervals continued to play a vital role in refining hydraulic fracturing design. They provided valuable insights into closure stress, leak-off behavior, and reservoir pressure, which were used to calibrate fracture models and optimize treatment parameters.

Well Testing and Production Analysis

Well Post-Frac Cleanup and Test Design Optimization

Well testing operations in the field were designed to measure all flowing parameters during post-frac well cleanup, in line with the philosophy that "what is measured can be optimized". A robust solids removal system consisting of a Sand Management System combined with a 4-phase separator was required to facilitate separator measurements while recovering significant back-produced proppant. An added efficiency of this arrangement is that well cleanup will transition seamlessly into well testing without the need to mobilize separate equipment. Practically it is observed that well behavior is established and basic well test parameters are often obtained even before solids cleanup is finished, meaning that a separate "well test" period is not required. In addition, the ability to cross-check well testing parameters from multiple separate measurement methods was useful to facilitate real-time identification and correction of operational errors, assuring the quality and proper management of operations.

Collaborative analysis of the full range of cleanup behavior quickly led to actionable insights into well behaviors and rapid improvement in the efficiency of flowback operations. Reliable trends were identified in fracture cleanup indicators such as:

- Water to Gas Ratio
- Condensate to Gas Ratio
- Flowing Tubing Head Pressure versus Gas Rate
- Water Salinity
- Total Frac Load Recovery

An example of an early trend in Basic Sediment and Water (BS&W) readings is described in Fig. 13 below. Well-A is an example of an ideal cleanup behavior of fracture fluid recovery from a single reservoir unit. Initial BS&W is high as frac water is exclusively unloaded prior to significant gas breakthrough, later transitioning to increased hydrocarbon production and a low formation water rate. In Well-C the most productive zones were already cleaned up in a prior intermediate flowback step. BS&W is initially very low and does not increase until higher drawdown is applied later in the choke schedule. Well-B exhibits mixed behavior since the zones which were previously cleaned up contributed a modest proportion of the overall productivity.

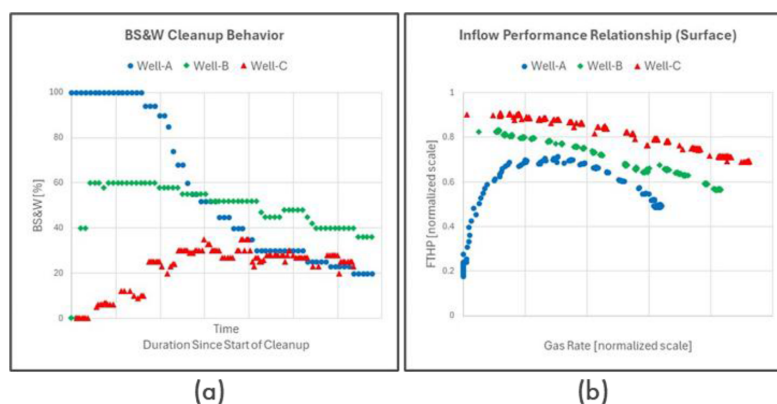


Figure 13—(a) BS&W and (b) surface IPR measurement trends used as proxies for fracture cleanup efficiency in three example wells. Each flowback utilized a similar time-based bean-up schedule and differences in cleanup behavior are attributed to aspects of the well and completion.

Considered together with other well test parameters, this data highlighted a significant opportunity for optimization of early well unloading design using relatively simplistic data. It was identified that some wells would exhibit delayed fracture cleanup using a fixed bean-up schedule due to initial dominance of zones which were not significantly loaded with fracture fluids or highly productive layers, either due to a previous intermediate clean-up step or due to being partially completed without fracturing. A revised accelerated early bean-up plan was designed for wells with this completion plan resulting in a 24% reduction in overall cleanup durations with associated cost savings.

The duration of well tests is often the subject of debate, posing a potential conflict between extensive testing strategies with drawdown management and more aggressive flowback plans which minimize time, cost, and consumption of resources. The objective of collaborative data analysis is to create a shared understanding and vocabulary across functional teams allowing identification and prediction of the point of diminishing returns during well cleanup and testing, facilitating high quality informed decisions with cross-discipline alignment. Data-driven improvements in execution efficiency as described above, as well as competitive scoping while transitioning away from appraisal and into field development, has led to a significant reduction of 64% in overall flowback durations from Campaigns 1 to 3 as shown in Fig. 14, while continuing to deliver both well cleanup and well testing objectives. Average WGRs by the end of

the flowback are also plotted for each campaign to emphasize sufficient clean-up was maintained without compromising the quality of the test data.

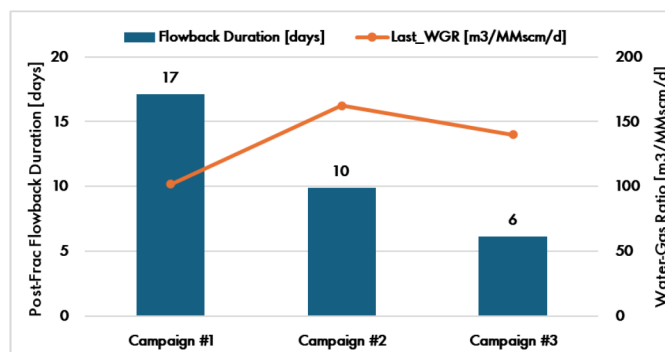


Figure 14—Comparison of Post-Frac flowback duration and the Water-Gas Ratio by the end of the flowback across the 3 well campaigns indicating a progressive reduction in flowback time by 64% from Campaign #1 to Campaign #3 ending the test at a reasonable WGR that is expected to decline further once the well is put on-stream

PLT Optimization

Production logging in the field is used to allocate zonal contribution and evaluate productivity as well as identify water influx points and is often a critical input to reduce reservoir uncertainties. Initial production logs in the field were designed with flowing passes at three different flow rates to characterize inflow under varying drawdowns (High, Medium, and Low). Post-job analysis quickly confirmed that relative zonal contribution did not vary significantly at the different flow rates. This is attributed to the relatively similar reservoir pressure regimes and minimal reservoir pressure depletion.

Job design was updated to perform logging passes at two flow rates only (High and Low), and to incorporate a decision point in the QC process to trigger logging at a third flow rate only if significant deviation from such ideal behavior was identified. This optimization was supported by use of electric-line logging with surface read out, allowing analysis of logging passes prior to proceeding. By eliminating one stabilization and logging period the overall job duration was significantly reduced.

Flowrate selection was also optimized by taking into account the well dynamics and flow regime within the wellbore, which facilitates selecting the widest range of measurement points while reducing the risk of extended stabilization periods or failure to achieve proper downhole stability due to the onset of liquid loading. An example of this exercise is shown in Fig. 15 below highlighting the conditions within the well performance curve at which liquid loading is un/likely to occur.

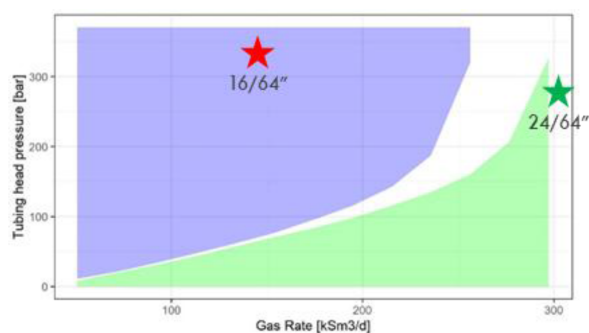


Figure 15—Example of a liquid loading regime plot for a given well geometry. The blue area represents the region of likely liquid loading while green indicates a more stable lifting regime, with white indicating a transition area. The well-specific productivity curve from well test data is mapped to determine the expected choke setting to achieve the desired flow regime in the wellbore during PLT logging.

Furthermore, multiple logging passes are usually done at each selected flow rate to help in calibrating spinner response and interpreting the zonal contribution. The choice of line speed should be suitable for flow velocities and conditions to be measured. Upon re-evaluating the initial job designs, it was determined that line speeds could be increased significantly, which would reduce the actual acquisition time by approximately 50%, contributing to more efficient execution.

As job experience was gained and log acquisition time was reduced, the opportunity for reducing data transfer time for decision making and QA/QC became more apparent. Stabilization criteria were reviewed by back testing existing data and incorporated into well programs as clear go/no-go criteria to commence logging passes. In cases where the stability criteria are not met within a pre-determined time period a data review is triggered to evaluate the rate of change in trends and decide the way forward. This transferred the primary confirmation to commence logging away from the office and to the wellsite in cases where data review is not required, streamlining a significant communication step.

These optimizations together have proven to reduce the duration of PLT flowing surveys by an average of 50% as shown Fig. 16 below, with an associated reduction in time-related costs.

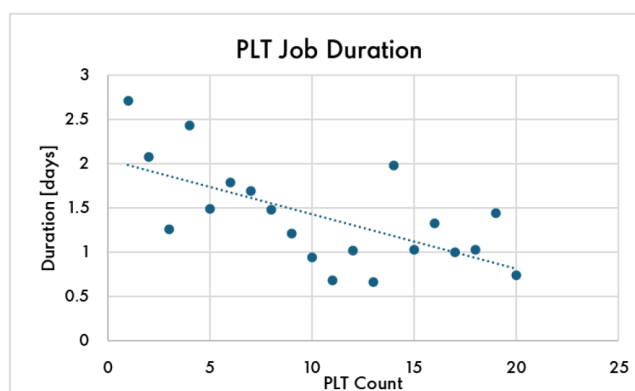


Figure 16—Average PLT flowing duration was reduced by half by optimizing job design, logging speeds, and stabilization criteria. Swifter decisions may be made in field by clearly communicating go/no-go criteria for logging and triggers for specialist quality assurance and decision-making.

Discussion

The integrated data acquisition and analysis approach applied in this study has demonstrated significant value in optimizing hydraulic fracturing and well testing in tight gas-condensate reservoirs. The iterative workflow, which combined static and dynamic data sources with real-time diagnostics, enabled continuous refinement of fracture designs and operational strategies across multiple well campaigns.

One of the key insights from the campaign was the shift from a volume-driven to a value-driven stimulation strategy. The reduction in proppant intensity and fluid volume per stage, without compromising well productivity, underscores the importance of fracture efficiency over total material input. This was further validated by PTA results, which showed no significant improvement in fracture conductivity with higher proppant tonnage when comparing wells of similar rock quality, and by tracer logs confirming effective fracture coverage even with fewer stages. However, it is also worth noting that these observations focus on a relatively short-term production performance within early well-life and long-term consequences on hydrocarbon recovery require further verification after sufficient production history from these wells.

Furthermore, the optimization of fracture stage count, particularly in the Miqrar reservoir, highlighted the benefits of leveraging diagnostic tools such as radioactive tracers and fracture modeling. These tools revealed that fewer, more effectively placed stages could achieve sufficient reservoir contact, reducing completion complexity and cost. The ability to contain fracture growth within target zones also minimized water production risks, enhancing long-term well performance.

DFIT analysis played a dual role in both informing fracture design and identifying operational inefficiencies. The recognition of stage-to-stage communication and its impact on DFIT reliability led to a strategic reduction in test frequency, saving time and resources while maintaining data quality through selective, representative testing.

Well testing and flowback operations also benefited from a data-driven approach. The integration of cleanup behavior indicators and surface IPR trends enabled the development of accelerated bean-up schedules, reducing cleanup durations by 24% and overall well flowback by up to 64%. Similarly, PLT job design was optimized through real-time decision-making and flow regime analysis, cutting survey durations by 50% without compromising data integrity.

Collectively, these findings demonstrate that a holistic, diagnostics-informed approach can significantly enhance both technical and commercial outcomes in tight gas-condensate developments. The continuous feedback loop between data acquisition, analysis, and design refinement was instrumental in achieving these results.

Conclusion

This study presents a successful case of leveraging integrated data acquisition and diagnostic analysis to optimize hydraulic fracturing and well testing in a tight gas-condensate greenfield in North Central Oman. The key conclusions are:

- **Fracture Efficiency Over Volume:** Optimization efforts demonstrated that reducing proppant and fluid volumes per stage, while maintaining or improving gas rates, is achievable through data-informed design adjustments. This shift from volume-driven to efficiency-driven stimulation led to a 49% improvement in gas rate per proppant tonnage.
- **Stage Count Optimization:** Reducing the number of fracture stages per well, supported by tracer diagnostics and fracture modeling, maintained effective reservoir contact while lowering operational complexity and cost.
- **Selective DFIT Application:** Identifying non-representative DFIT conditions due to stage communication allowed for the strategic elimination of redundant tests, improving operational efficiency without compromising model calibration.
- **Well Testing and PLT Optimization:** Streamlined post-frac flowback and PLT jobs, enabled by real-time diagnostics and collaborative analysis, reduced operation duration by 64% and 50% respectively, while preserving data quality and test objectives.
- **Sustainable Development:** The approach reduced material usage, operational time, and costs—contributing to more sustainable field development. These efficiencies support lower environmental impact and improved resource recovery.

The methodology outlined in this paper offers a robust framework for future developments in similar tight gas-condensate environments. Continued application and refinement of this integrated, data-driven approach will be essential for maximizing value and ensuring long-term production sustainability.

Further insights and learnings from the primary completions phase, represented by the three campaigns discussed in this paper, indicate there still remains room for further optimization. While the current work focused on key areas of hydraulic fracturing and well testing, additional opportunities exist to enhance performance and efficiency. Notably, the perforation strategy requires re-evaluation to mitigate solids production issues observed in several wells. This topic, along with other critical elements such as frac fluid selection and solids management remain areas of active investigation. These aspects are expected to play a significant role in further enhancing well performance and will be addressed in future publications. Their evaluation is essential to fully unlock the potential of tight gas-condensate reservoirs and to ensure long-term, sustainable field development. This is especially important as the asset matures and reservoir pressure

depletes, which will require further optimization of hydraulic fracturing designs to maintain productivity and economic viability.

Abbreviations

SDO:	Shell Development Oman
PVT:	Pressure-Volume-Temperature
GOR:	Gas-Oil Ratio
CGR:	Condensate-Gas Ratio
DFIT:	Diagnostic Fracture Injection Test
PTA:	Pressure Transient Analysis
PBU:	Pressure Build-Up
PLT:	Production Logging Test
BS&W:	Basic Sediment & Water
PQ:	Pressure vs Rate
IPR:	Inflow Performance Relationship
EGC:	Equivalent Gas Column
PI:	Productivity Index

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